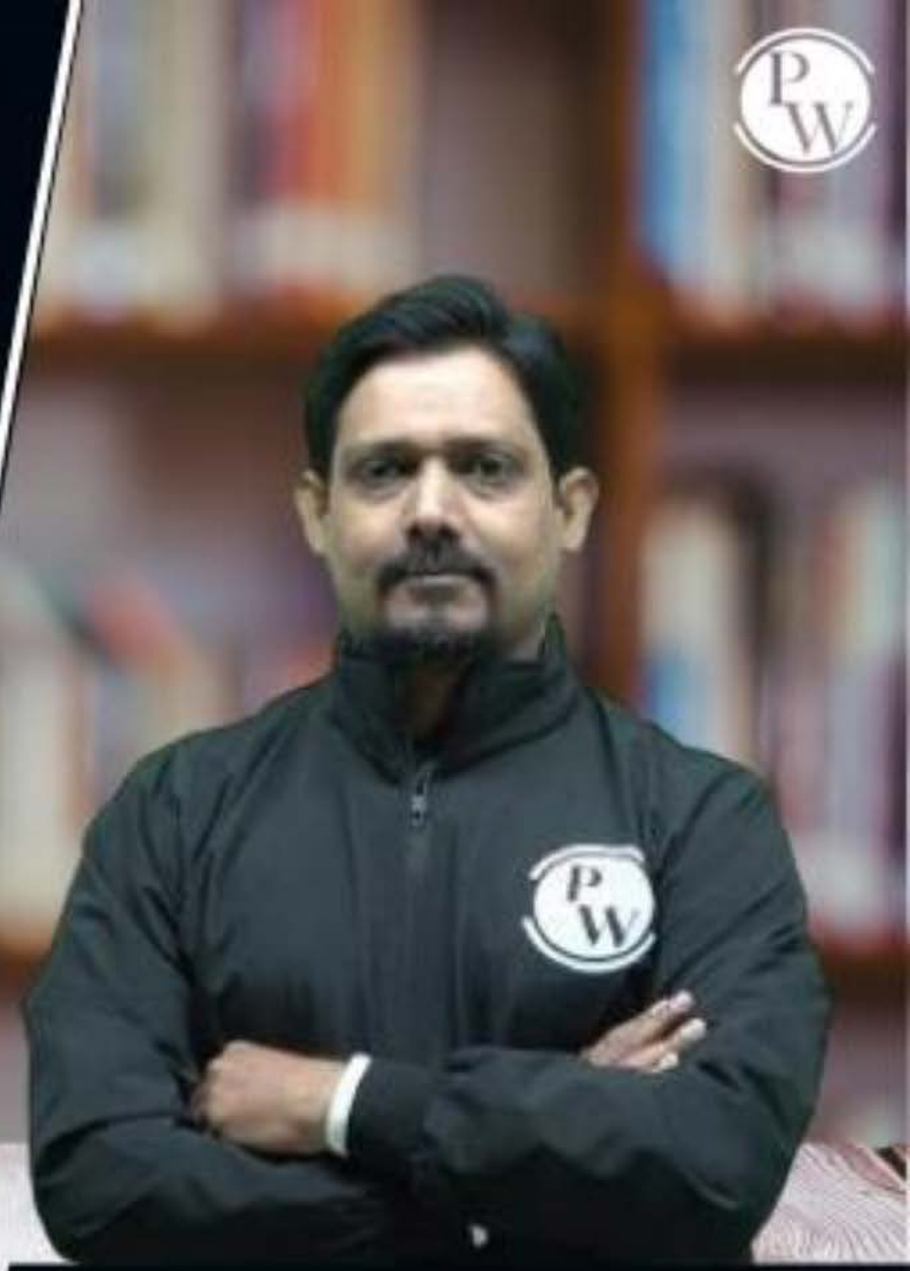
Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 05



By- Abhishek Sir

Recap of Previous Lecture



- Topic — Traversal
- Topic — Binary Search Tree
- Topic — Inorder traversal
- Topic
- Topic

Slide

Topics to be Covered



- Topic — BST Traversal (special care)
- Topic — Search, Min, max, Delete
- Topic — Counting of BST
- Topic — AVL tree
- Topic

Slide



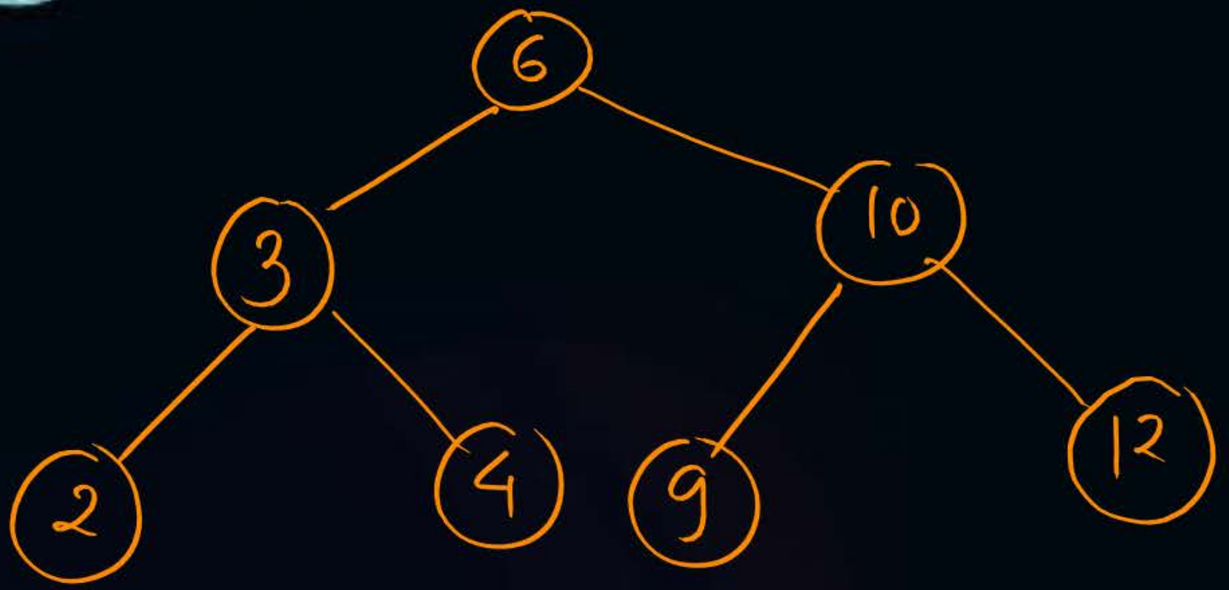
Topic : Tree

BST Preorder Post Order
BST Search , MIN MAX
BST DELETE
Counting
Problem Solving

Slide



Topic : Binary Search Tree



Inorder:

2, 3, 4, 6, 9, 10, 12

Increasing order (Sorted)

preorder :-

6, 3, 2, 4, 10, 9, 12

postorder

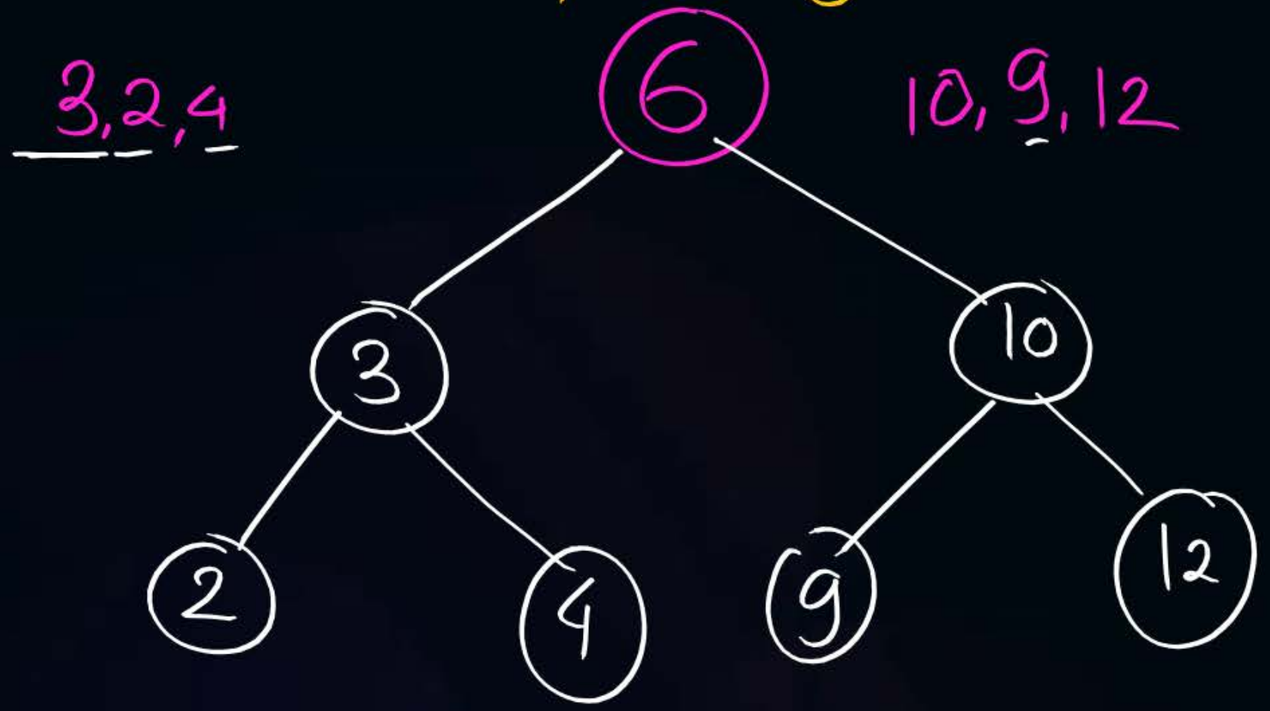
2, 4, 3, 9, 12, 10, 6

Slide



Topic : Binary Search Tree

Root Left Right



Binary Search preorder Tree is constructed

preorder :-

6, 3,2,4, 10,9,12

Left Right

Slide

Topic : Binary Search Tree

Left Right Root



2,4,3

9,12,10

Single postorder traversal

I can construct the BST

postorder

2, 4, 3, 9, 12, 10, 6

left Right

Slide



Topic : Binary Search Tree

post order of BST

1, 4, 7, 6, 3, 13, 14, 10, 8

$2^{5+3} + 2\times3 - 4\times5$

$= 256 + 6 - 20 = 242$

if Number of element in left subtree is x and Right subtree is y the $2^{x+y} + 2y - 4x$ is ____

Slide

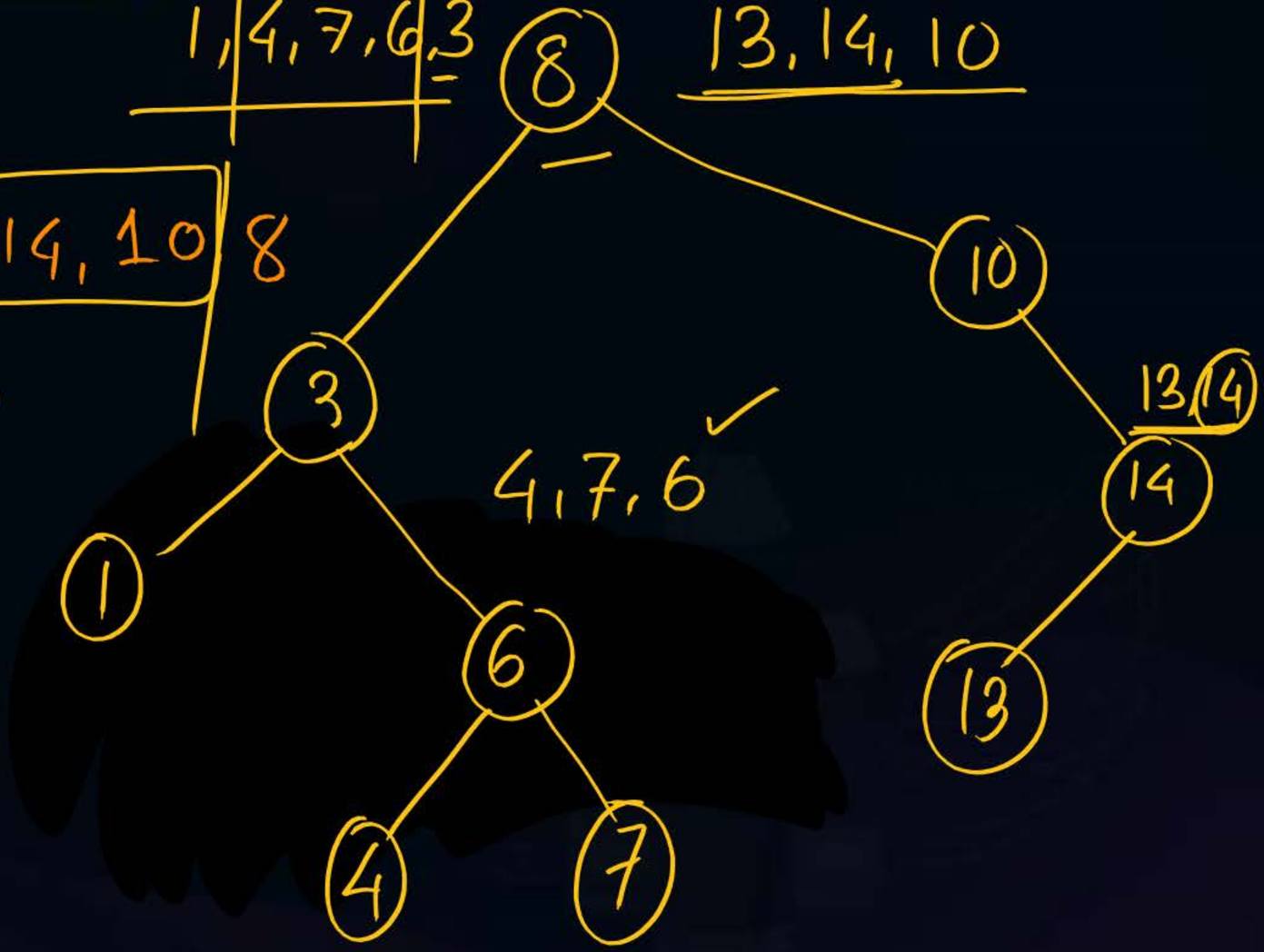


Topic : Binary Search Tree

post order of BST

1, 4, 7, 6, 3	13, 14, 10	8
5	3	

1, 4, 7, 6, 3 | 8 | 13, 14, 10



Slide



Topic : Binary Search Tree

if I say that a d b e f g h is postorder traversal of a full Binary tree. Can we construct the tree?

Yes

T



Two any tree with all Leaves at

Slide



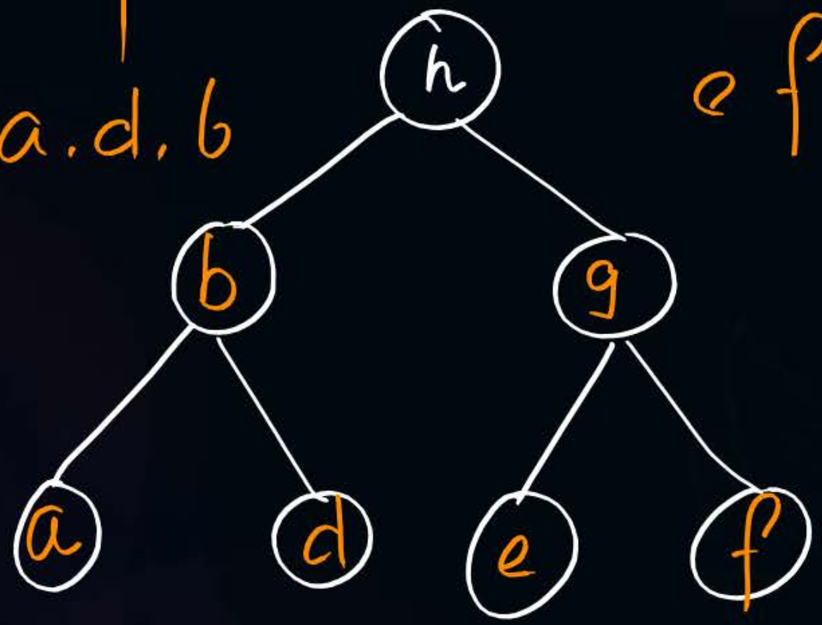
Topic : Binary Search Tree

3
7
15
31
63
127

$2^{h+1}-1$

a d, b, e, f, g, h postorder — full Binary tree : $2^{h+1}-1$

a . d . b e f . g



Slide



Topic : Binary Search Tree

1 Binary Tree : Inorder, preorder, tree can constructed

2. Binary Tree, Inorder, postorder, tree can be constructed.

3. FBT. Inorder — tree can be constructed
   preorder — tree can be constructed
   postorder — tree can be constructed

4 BST Inorder :- Can't be constructed
   preorder : Tree can be constructed.
   postorder : Tree can be constructed.

Slide



Topic : Binary Search Tree

5. 2 - any tree, preorder & postorder traversal we can construct tree uniquely. (*)

Slide

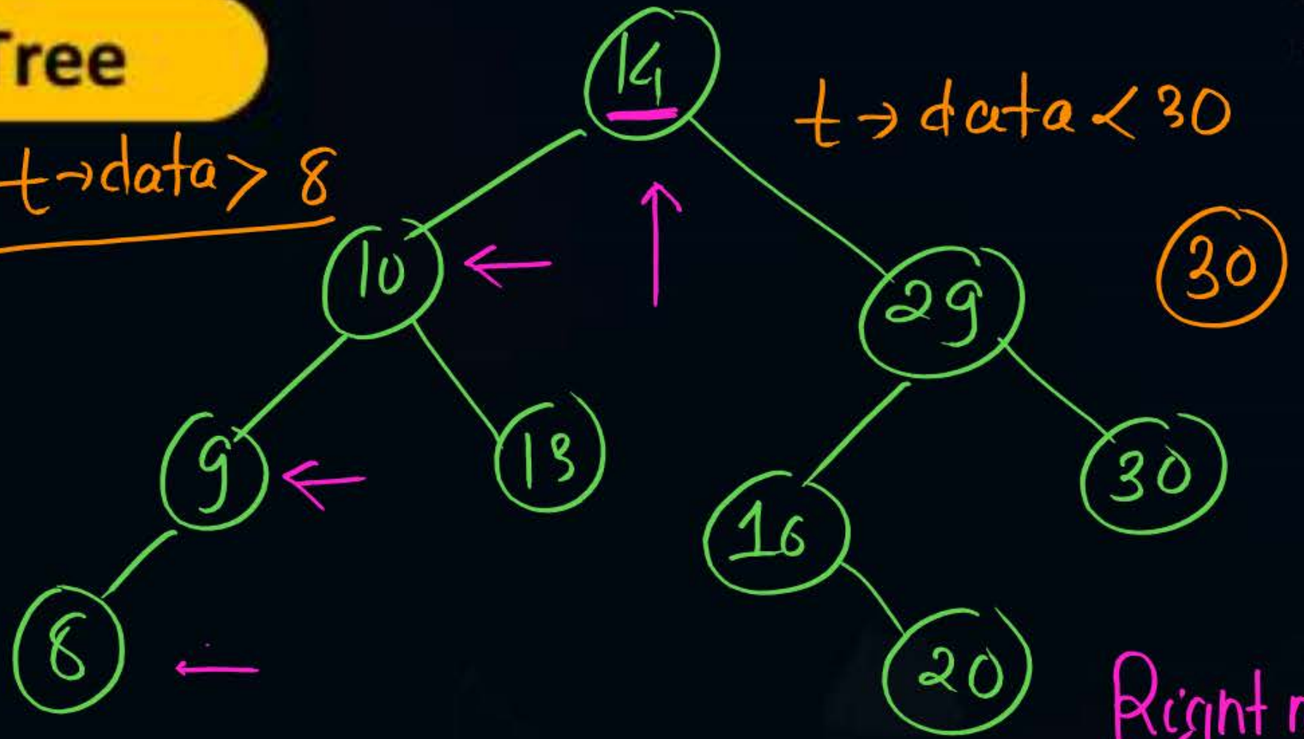


Topic : Binary Search Tree

Searching In BST

Minimum element

Left most Node



Right most Node is maximum

Slide



Topic : Binary Search Tree

```
int Search ( BTnode *t, int x){
  if (t != NULL) {
      if (t->data == x)
          return 1;
      if (t->data > x)
          return Search (t->Left, x);
      else
          return Search (t->right, x);
  }else
      return -1;
}
```

```
typedef struct btnode {
    int data;
    struct btnode *left;
    struct btnode *right
} BTnode;
```

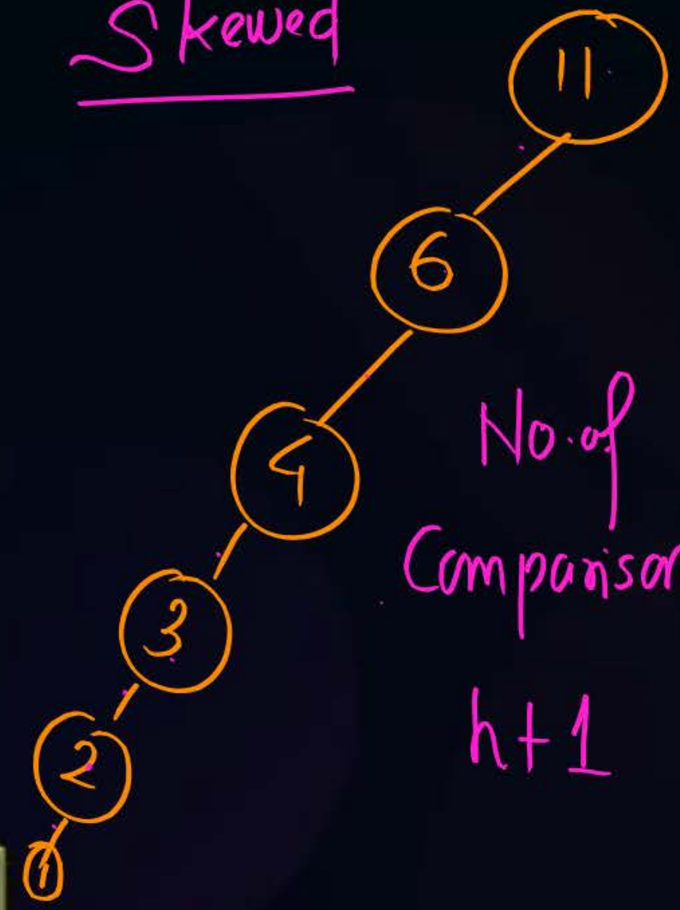
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Topic : Binary Search Tree

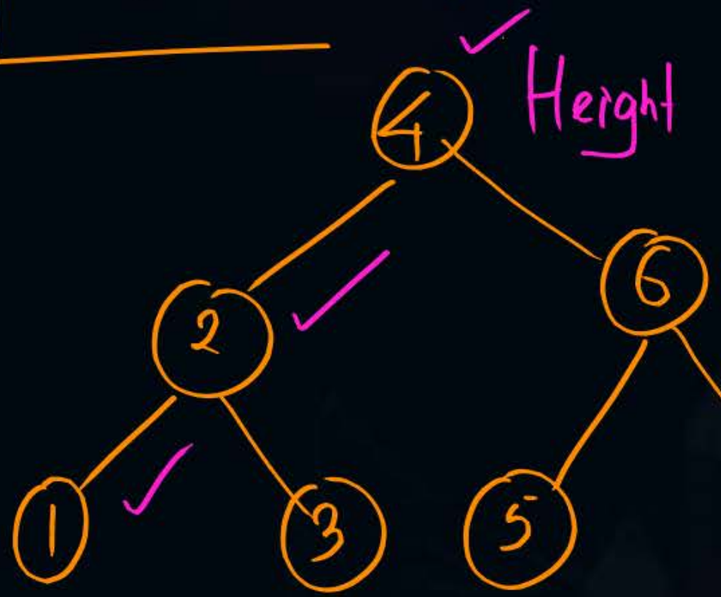
Time Complexity of Search operation in BST

Skewed



No. of Comparison

$h+1$



Height balanced

$\lfloor \log_2 7 \rfloor + 1$

No of comparison

$\lfloor \log_2 n \rfloor$

Slide



Topic : Binary Search Tree

Important

maximum

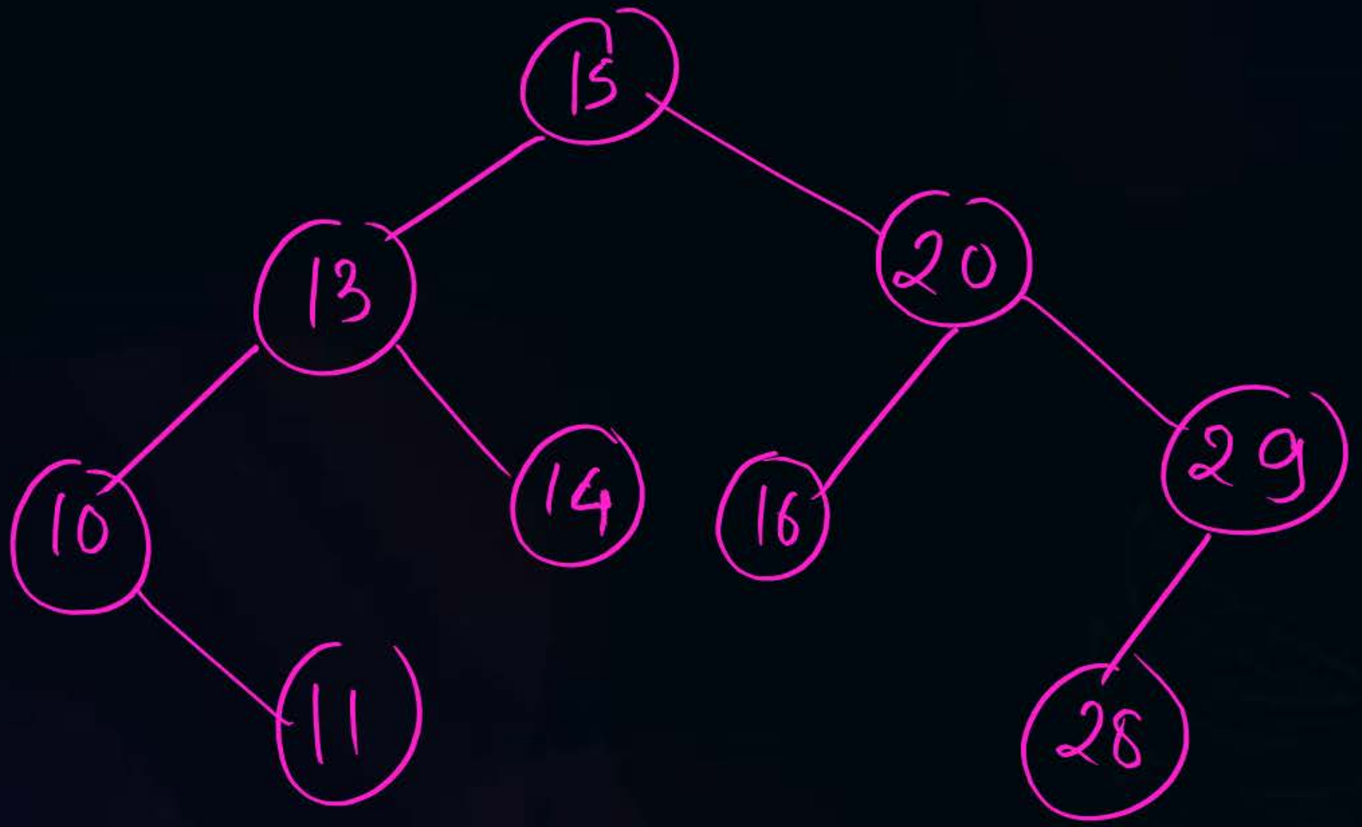
O(h) — skewed — O(n)

O(h) — balanced :- O($\log_2 n$)

Slide



Topic : Binary Search Tree



Slide



Topic : Binary Search Tree

Min/Max

Complexity

O(h)

```
int min (BTnode *t){
    if(t!=NULL){
        while(t->left!=NULL)
            t=t->left;
    } return t->data;
    else{
        printf("Empty tree");
    } return -1
}
```

```
int max(BTnode *t){
    if(t!=NULL){
        while(t->right!=NULL)
            t=t->right;
        return
            t->data;
    }else{
        printf("Empty tree")
    } { return -1
}
```

Slide



Topic : Binary Search Tree

O(h) → Skewed O(n)

balanced O(log₂n)

Balanced BST : O(log₂n)

BST — worstcase heigh

Linear

Slide



Topic : Binary Search Tree

Insert, min, max, Search $\longrightarrow O(h)$

10:39

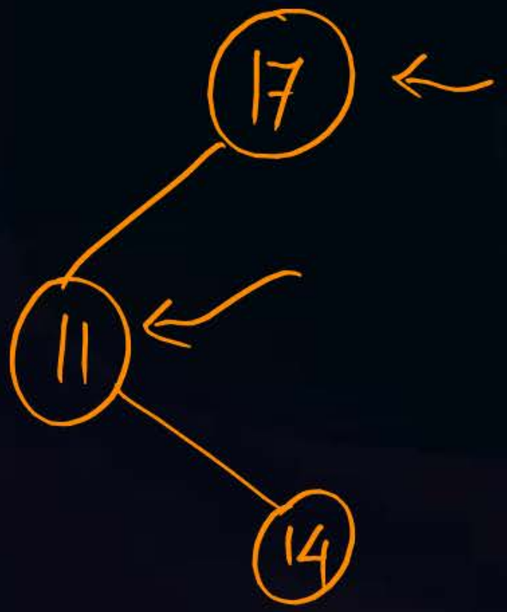
Slide



Topic : Binary Search Tree

#Q if an unlabelled tree with n node is given & n distinct value is given.. How many ways we can fill the unlabelled tree so that it will become a BST.

1 way we can fill



11, 14, 17

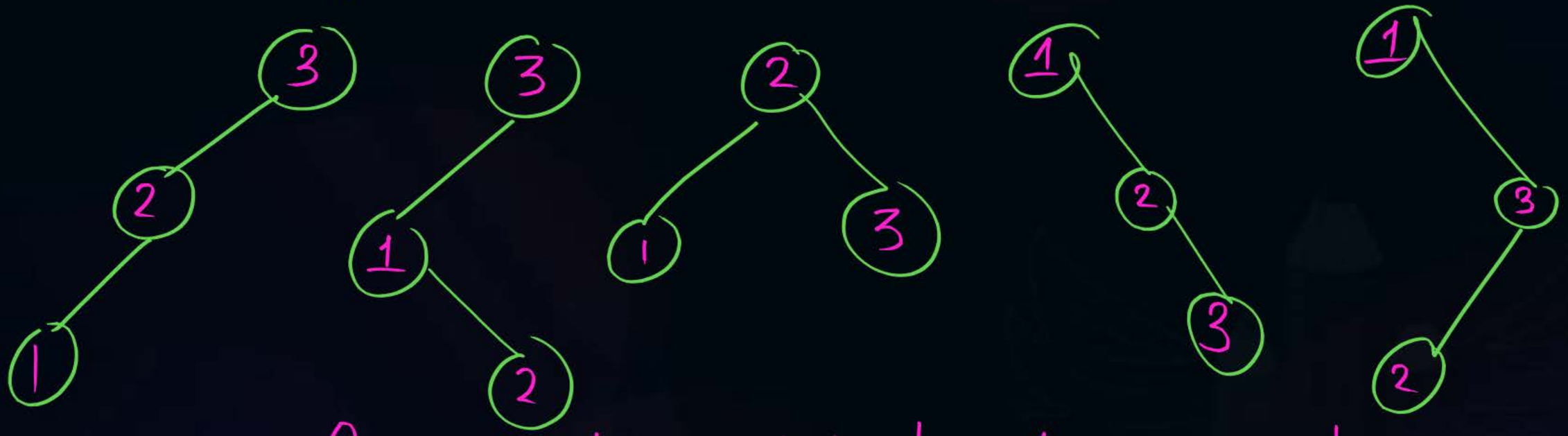


Slide



Topic : Binary Search Tree

No. of unlabelled trees with 3 node (1,2,3)



No. of BST with n distinct node is $\frac{1}{n+1} {}^{2n}C_n$

Slide



Topic : Binary Search Tree

#Q if BST constructed with key value

4, 9, 20, 21, 26, 30, 91, 80, 85, 89, 100, 120

with 80 as Root of BST. The No. of BST possible is _____

Slide

Topic : Binary Search Tree

$11 \times 12 = 132$

$\frac{1}{7}\frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$

#Q if BST constructed with key value

4, 9, 20, 21, 26, 30, 91, 80, 85, 89, 100, 120

$C_6 = \frac{1}{7}{}^{12}C_6$

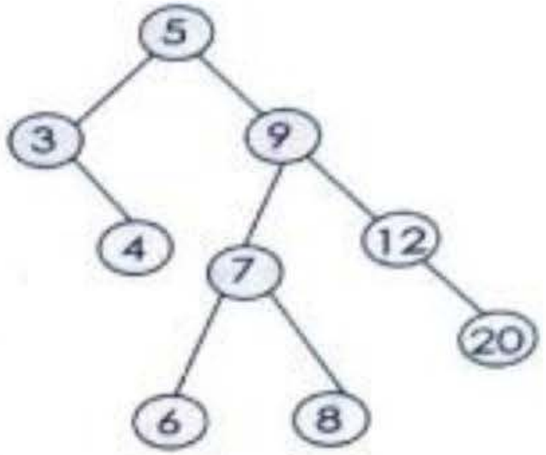


Slide



Topic : Binary Search Tree

32. The binary search tree shown below was constructed by inserting a sequence of items into an empty tree.



Which of the following input sequences will produce this binary search tree?

(a) 5 3 4 9 12 7 8 6 20
(b) 5 9 3 7 6 8 4 12 20
(c) 5 9 7 3 8 12 6 4 20
(d) 5 9 3 6 7 8 4 12 20

Slide



Topic : Binary Search Tree

31. The Preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42.

Which one of the following is the not Post order traversal sequence of the same tree?

(a) 10, 20, 15, 23, 25, 35, 42, 39, 30
(b) 15, 10, 25, 23, 20, 42, 35, 39, 30
(c) 15, 20, 10, 23, 25, 42, 35, 39, 30
(d) 15, 10, 23, 25, 20, 35, 42, 39, 30

preorder
30, 20, 10, 15, 25, 23, 39, 35, 42

Not postorder

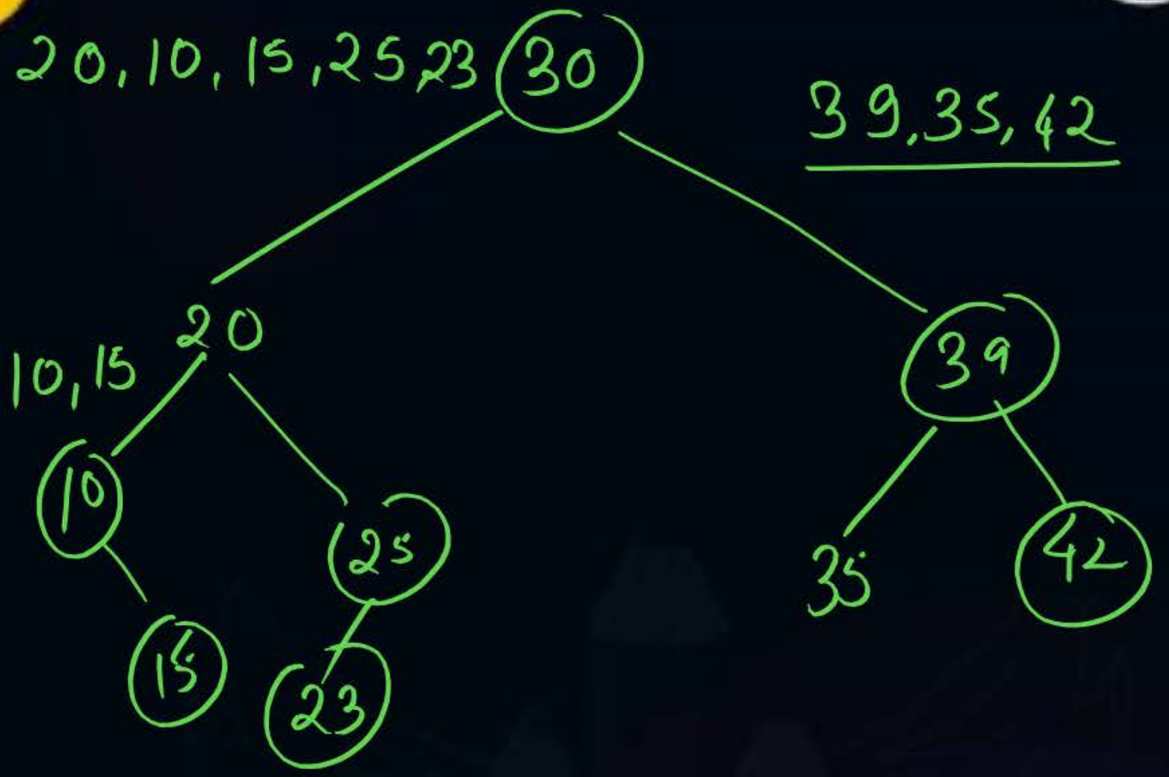
(a) 10, 20, 15, 23, 25, 35, 42, 39, 30
(b) 15, 10, 25, 23, 20, 42, 35, 39, 30
(c) 15, 20, 10, 23, 25, 42, 35, 39, 30
(d) 15, 10, 23, 25, 20, 35, 42, 39, 30

Topic : Binary Search Tree

31. The Preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42.

Which one of the following is the not Post order traversal sequence of the same tree?

(a) 10, 20, 15, 23, 25, 35, 42, 39, 30 ✗
(b) 15, 10, 25, 23, 20, 42, 35, 39, 30 ✗
(c) 15, 20, 10, 23, 25, 42, 35, 39, 30
(d) 15, 10, 23, 25, 20, 35, 42, 39, 30



15, 10, 23, 25, 20, 35, 42, 39, 30

Slide



2 mins Summary

Topic — Traversal special case

Topic — Search, min max

Topic — Counting of BST

Topic

Topic

Slide

THANK - YOU